

LayoutManagerDemo

Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class LayoutManagerDemo extends JFrame implements ActionListener {

    JPanel mainPanel;
    JRadioButton flowButton, borderButton, gridButton;
    ButtonGroup group;

    LayoutManagerDemo() {

        setTitle("Layout Manager Demo");
        setSize(500, 350);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        // Radio buttons
        flowButton = new JRadioButton("FlowLayout", true);
        borderButton = new JRadioButton("BorderLayout");
        gridButton = new JRadioButton("GridLayout");

        group = new ButtonGroup();
        group.add(flowButton);
        group.add(borderButton);
        group.add(gridButton);

        JPanel topPanel = new JPanel();
        topPanel.add(flowButton);
        topPanel.add(borderButton);
        topPanel.add(gridButton);

        // Register listeners
        flowButton.addActionListener(this);
        borderButton.addActionListener(this);
        gridButton.addActionListener(this);

        // Main panel
        mainPanel = new JPanel();
        mainPanel.setLayout(new FlowLayout());

        createFlowLayout();

        add(topPanel, BorderLayout.NORTH);
        add(mainPanel, BorderLayout.CENTER);

        setVisible(true);
    }

    // FlowLayout
```

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void createFlowLayout() {
    mainPanel.removeAll();
    mainPanel.setLayout(new FlowLayout());

    mainPanel.add(new JButton("Button 1"));
    mainPanel.add(new JButton("Button 2"));
    mainPanel.add(new JButton("Button 3"));
    mainPanel.add(new JButton("Button 4"));
    mainPanel.add(new JButton("Button 5"));

    refreshPanel();
}

// BorderLayout
void createBorderLayout() {
    mainPanel.removeAll();
    mainPanel.setLayout(new BorderLayout(10, 10));

    mainPanel.add(new JButton("NORTH"), BorderLayout.NORTH);
    mainPanel.add(new JButton("SOUTH"), BorderLayout.SOUTH);
    mainPanel.add(new JButton("EAST"), BorderLayout.EAST);
    mainPanel.add(new JButton("WEST"), BorderLayout.WEST);
    mainPanel.add(new JButton("CENTER"), BorderLayout.CENTER);

    refreshPanel();
}

// GridLayout
void createGridLayout() {
    mainPanel.removeAll();
    mainPanel.setLayout(new GridLayout(2, 3, 10, 10));

    mainPanel.add(new JButton("Button 1"));
    mainPanel.add(new JButton("Button 2"));
    mainPanel.add(new JButton("Button 3"));
    mainPanel.add(new JButton("Button 4"));
    mainPanel.add(new JButton("Button 5"));
    mainPanel.add(new JButton("Button 6"));

    refreshPanel();
}

// Refresh panel
void refreshPanel() {
    mainPanel.revalidate();
    mainPanel.repaint();
}

// Handle radio button events
public void actionPerformed(ActionEvent e) {

    if (flowButton.isSelected()) {
        createFlowLayout();
    }
    else if (borderButton.isSelected()) {
        createBorderLayout();
    }
}

```

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        else if (gridButton.isSelected()) {
            createGridLayout();
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            new LayoutManagerDemo();
        });
    }
}

```

Output:



